

Kramers–Kronig Relations in Gravitational-Wave Lensing

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- TS, ApJ 896, (2020) (arXiv:2003.11748 [gr-qc])
- S.Tanaka and TS, PRD 108, 044015 (2023) (arXiv:2303.05650[gr-qc])
- S.Tanaka, G.Prabhu, S.Kapadia, TS, e-Print: 2504.21320 [gr-qc] (accepted for PRD)
- TS and S.Kapadia, PRD 112, 063529 (2025) (arXiv: 2506.02430 [gr-qc])

Gravitational lensing (GL) of light

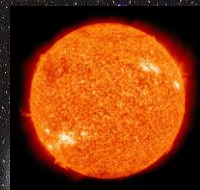
powerful for probing

- dark matter
- cosmological parameters
- growth of large-scale structure
- gravitational physics
- . . .

GWs are also lensed



©LIGO



Lensing object

GW



Difference between GL of light and GWs

e.g. Takahashi&Nakamura 2003

- **Coherent GW waveform**
- **Wave effects** $\lambda_{\text{light}} \sim 5 \times 10^{-7} \text{ m}$, $\lambda_{\text{GW}} \sim \text{3000 km}$

Additional probe of cosmological parameters

More information of lensing objects

New probe of sub-galactic-scale DM distribution

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A new frontier in cosmology!

No robust detection yet

GWTC-4.0: Searches for Gravitational-Wave Lensing Signatures

ABSTRACT

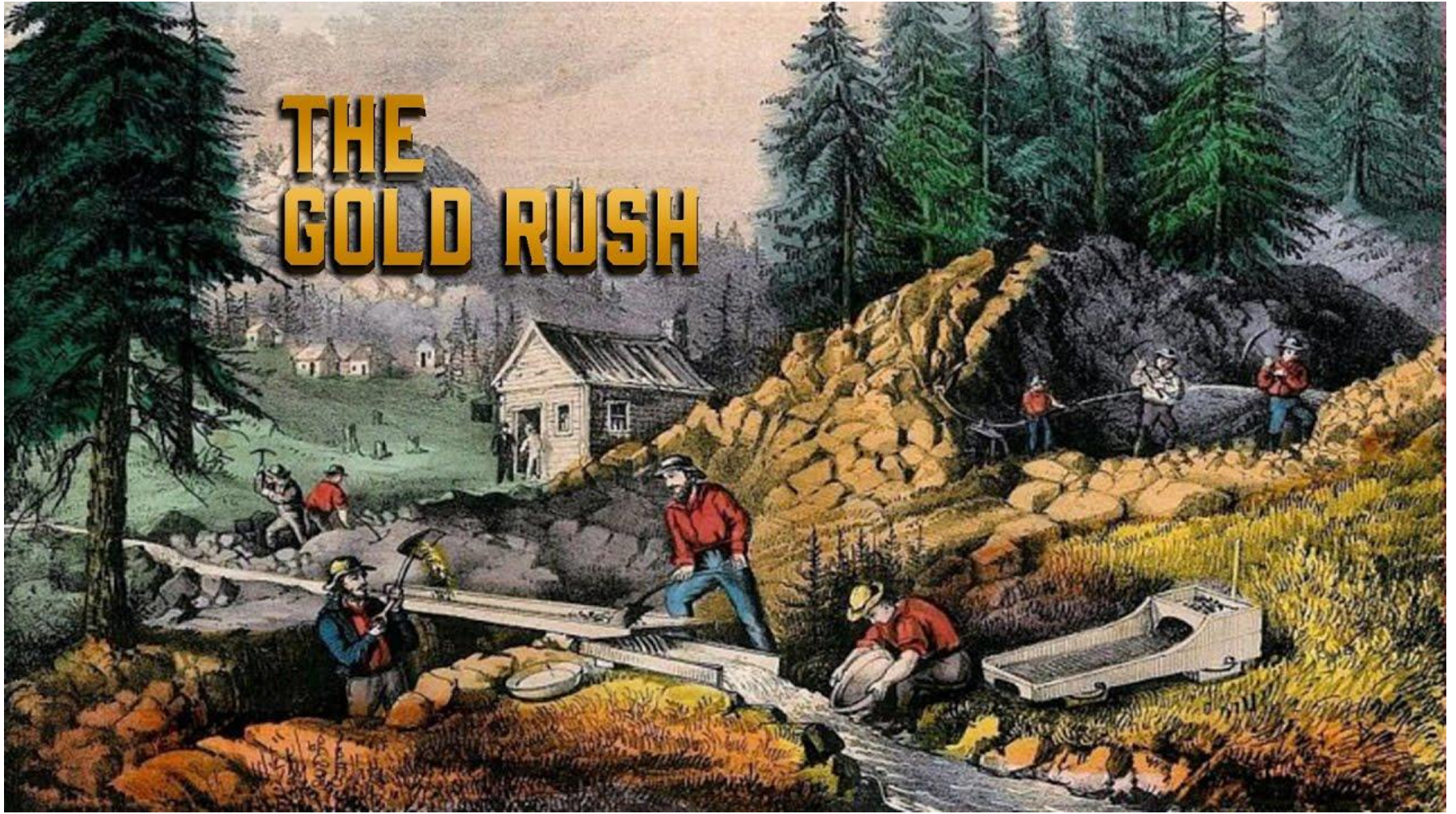
Gravitational waves can be gravitationally lensed by massive objects along their path. Depending on the lens mass and the lens–source geometry, this can lead to the observation of a single distorted signal or multiple repeated events with the same frequency evolution. We present the results for gravitational-wave lensing searches on the data from the first part of the fourth LIGO–Virgo–KAGRA observing run (O4a). We search for strongly lensed events in the newly acquired data by (1) searching for an overall phase shift present in an image formed at a saddle point of the lens potential, (2) looking for pairs of detected candidates with consistent frequency evolution, and (3) identifying sub-threshold counterpart candidates to the detected signals. Beyond strong lensing, we also look for lensing-induced distortions in all detected signals using an isolated point-mass model. We do not find evidence for strongly lensed gravitational-wave signals and use this result to constrain the rate of detectable strongly lensed events and the merger rate density of binary black holes at high redshift. In the search for single distorted lensed signals, we find one outlier: GW231123_135430, for which we report more detailed investigations. While this event is interesting, the associated waveform uncertainties make its interpretation complicated, and future observations of the populations of binary black holes and of gravitational lenses will help determine the probability that this event could be lensed.

arXiv:2512.16347

observatory/model	R_{obs} [yr^{-1}]	R_{sl} [yr^{-1}]	$\langle\mu_{\text{sl}}\rangle$	R_{pair} [yr^{-1}]	Δt [day]	$\langle\mu_{\text{leading}}/\mu_{\text{trailing}}\rangle$
aLIGO/Pop-I/II	1.14e+03	5.84e−01	14.35 (3.39–72.71)	7.77e−02	0.006 (0.000–0.739)	1.00 (0.61–1.23)
aLIGO/Pop-III (B17)	2.00e−01	6.21e−05	—	—	—	—
aLIGO/Pop-III (K16)	1.68e+02	3.89e−02	6.32 (2.50–27.97)	3.33e−03	0.433 (0.013–2.906)	1.22 (0.82–1.37)
aLIGO/PBH	4.75e+02	1.35e−01	6.89 (2.40–32.84)	1.43e−02	0.124 (0.002–2.853)	0.92 (0.48–1.54)
KAGRA/Pop-I/II	6.84e+02	1.69e−01	17.49 (3.30–105.11)	2.37e−02	0.002 (0.000–0.090)	1.00 (0.52–1.19)
KAGRA/Pop-III (B17)	5.58e−02	3.81e−06	—	—	—	—
KAGRA/Pop-III (K16)	4.59e+01	3.10e−03	7.65 (2.51–83.11)	6.67e−04	0.005 (0.002–0.008)	1.01 (1.00–1.01)
KAGRA/PBH	1.93e+02	2.00e−02	7.27 (2.65–45.64)	3.33e−03	0.546 (0.139–1.081)	1.05 (0.81–1.79)
ET/Pop-I/II	5.54e+05	1.12e+03	2.10 (0.88–3.55)	4.56e+02	13.741 (1.184–83.138)	2.36 (0.91–6.75)
ET/Pop-III (B17)	5.96e+03	7.38e+01	2.41 (1.70–4.32)	1.50e+01	16.518 (0.736–79.897)	1.95 (0.70–5.10)
ET/Pop-III (K16)	1.13e+05	4.86e+02	2.10 (0.83–3.40)	1.74e+02	15.094 (1.328–96.548)	2.61 (0.93–6.91)
ET/PBH	2.27e+05	1.18e+03	2.25 (1.36–3.93)	3.55e+02	12.942 (1.042–80.279)	2.06 (0.80–5.60)
CE/Pop-I/II	7.31e+05	1.60e+03	1.88 (0.38–3.09)	8.36e+02	20.600 (2.318–113.044)	3.64 (1.24–11.20)
CE/Pop-III (B17)	1.54e+03	1.51e+01	2.44 (1.88–3.98)	2.60e+00	8.266 (0.501–208.184)	3.02 (1.02–6.55)
CE/Pop-III (K16)	9.96e+04	3.96e+02	2.07 (0.60–3.64)	1.82e+02	21.283 (1.444–107.229)	2.90 (0.92–8.78)
CE/PBH	2.47e+05	1.07e+03	2.05 (0.71–3.49)	4.63e+02	18.806 (1.290–108.130)	2.68 (1.01–8.18)
B-DECIGO/Pop-I/II	2.02e+05	4.71e+02	2.36 (1.63–4.19)	9.98e+01	8.252 (0.595–56.830)	1.70 (0.78–4.65)
B-DECIGO/Pop-III (B17)	5.96e+03	9.20e+01	2.50 (1.76–4.84)	1.92e+01	3.430 (0.188–21.441)	1.23 (0.50–2.82)
B-DECIGO/Pop-III (K16)	7.66e+04	3.86e+02	2.27 (1.47–3.94)	1.22e+02	14.577 (1.060–86.073)	1.88 (0.78–4.78)
B-DECIGO/PBH	1.31e+05	1.41e+03	2.63 (1.81–5.43)	2.70e+02	4.965 (0.264–50.640)	1.29 (0.57–3.29)

In the future, lensing of the GWs will become daily events.

THE GOLD RUSH

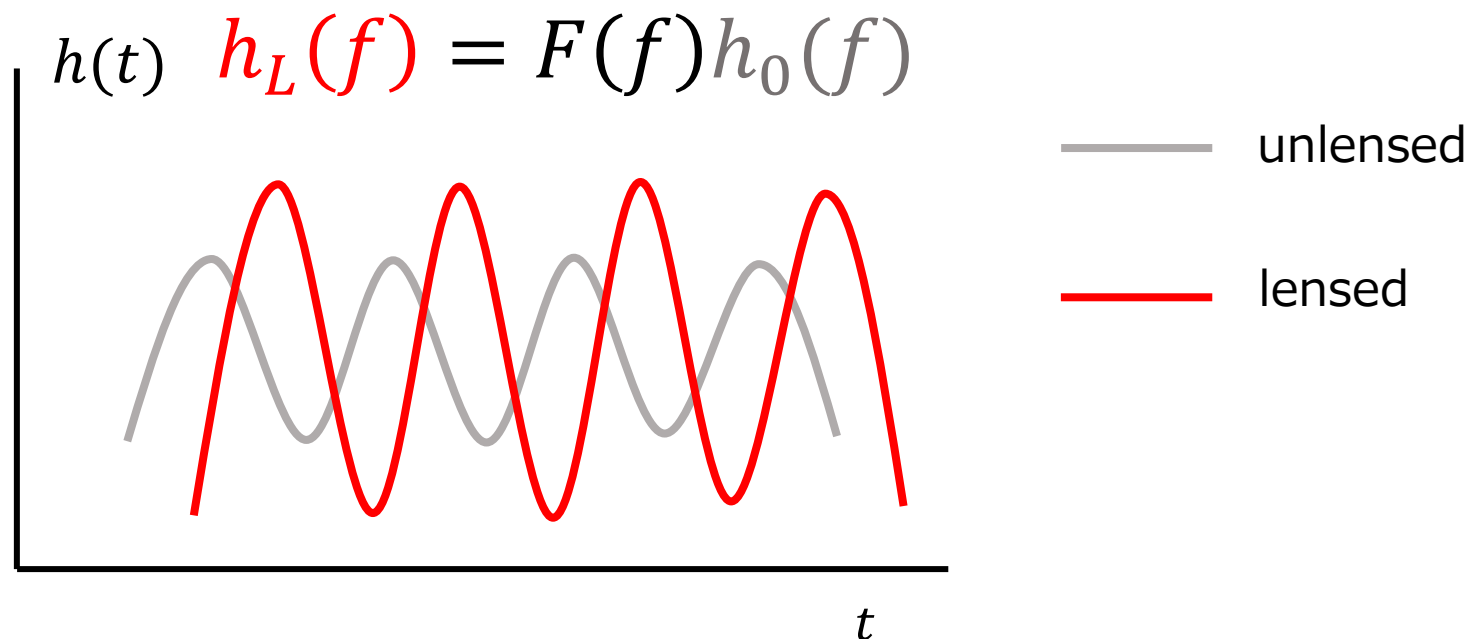


The golden era of the GL of GWs coming soon

Theoretical understandings in wave optics

Lensing of GWs

Lensing deforms the waveform in f-dependent manner

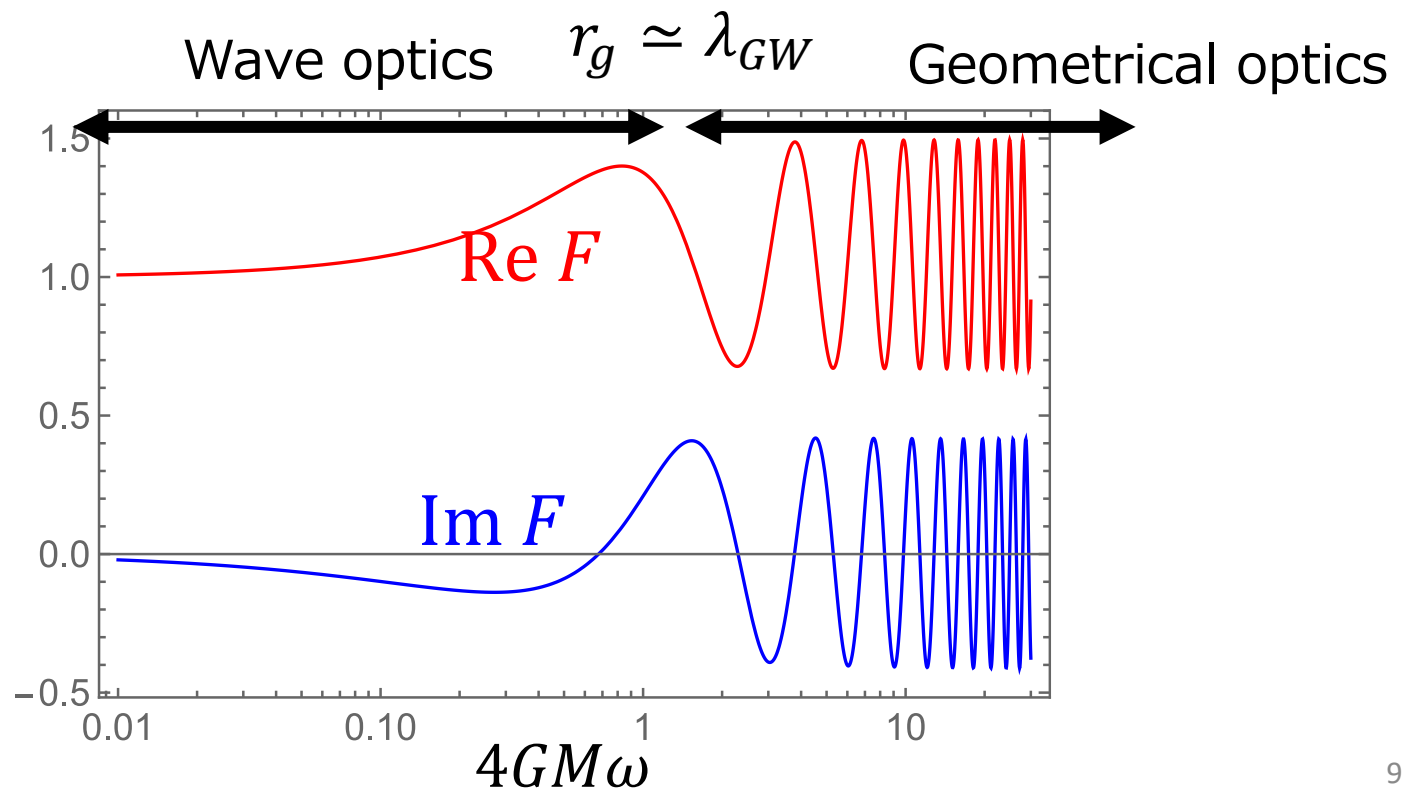
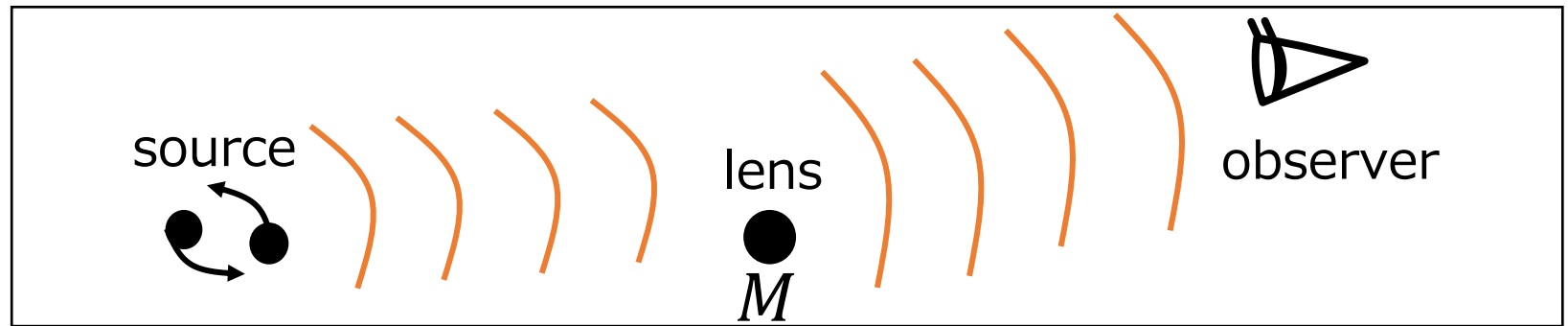


Amplification factor (complex number)

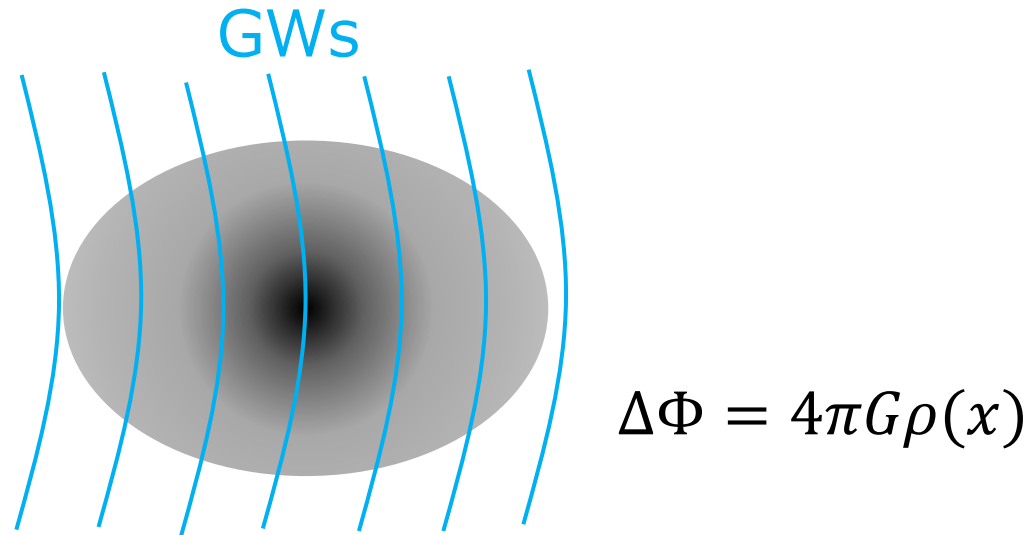
$$F(f) = A(f)e^{i\phi(f)}$$

Measurement of lens = measurement of $F(f)$

Example: point-mass lens



Wave optics



$$\bar{\nabla}^\alpha \bar{\nabla}_\alpha h_{\mu\nu} + 2\bar{R}_{\alpha\mu\beta\nu} h^{\mu\nu} = 0$$

Decompose $h_{\mu\nu}$ as $h_{\mu\nu} = \phi e_{\mu\nu}$ $e_{\mu\nu} \approx \text{const. during propagation}$

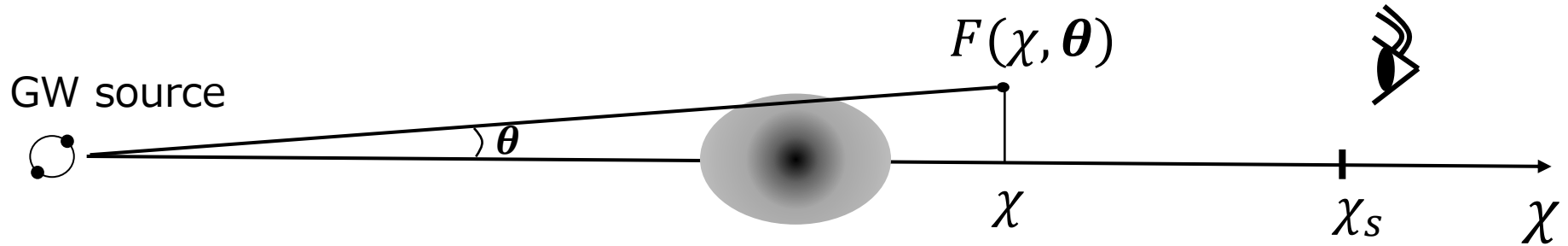
Frequency domain


$$(\nabla^2 + \omega^2)\phi(f, \mathbf{x}) = 4\omega^2\Phi(\mathbf{x})\phi(f, \mathbf{x})$$

Scattering is frequency dependent

Wave optics

Nakamura&Deguchi 1999



Small angle approximation ($|\theta| \ll 1$), $F_{,\chi} \ll \chi^{-1} F_{,\theta}$

$$(\nabla^2 + \omega^2)\phi(f, \mathbf{x}) = 4\omega^2 \Phi(\mathbf{x})\phi(f, \mathbf{x})$$

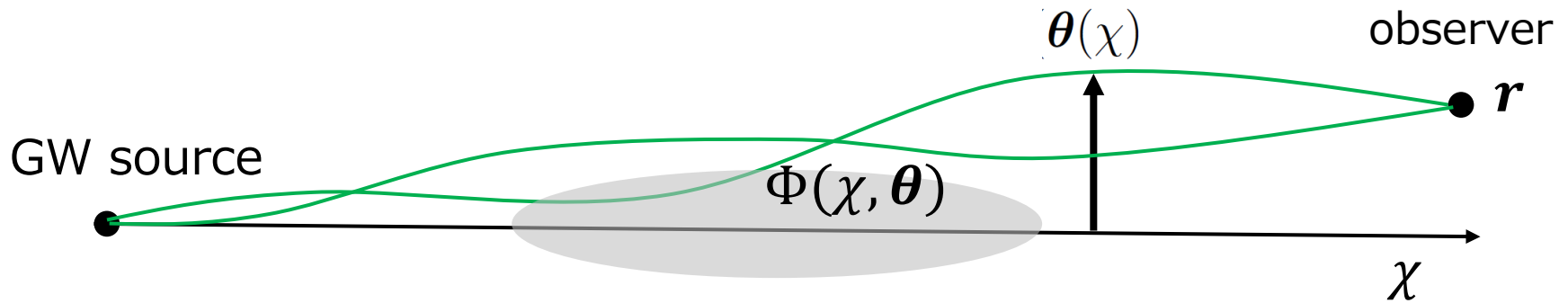
$$\Rightarrow i\omega^{-1} \frac{\partial F}{\partial \chi} = -\frac{1}{2\omega^2 \chi^2} \nabla_{\theta}^2 F + 2\Phi(\chi, \theta)F$$

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \nabla_{\theta}^2 F + VF \quad (\hbar \leftrightarrow \omega^{-1}, t \leftrightarrow \chi, m \leftrightarrow \chi^2, V \leftrightarrow 2\Phi)$$

Wave optics = 2d quantum mechanics with time-dependent mass and potential

Path integral formula

Nakamura&Deguchi 1999



$$F(\mathbf{r}, f) = \int D\theta \exp \left(i\omega \int_0^{\chi_s} L(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}}, \chi) d\chi \right)$$

$$\int_0^{\chi_s} L d\chi = t_d[\boldsymbol{\theta}] \quad t_d: \text{light travel time}$$

$$L(\boldsymbol{\theta}(\chi), \dot{\boldsymbol{\theta}}(\chi), \chi) = \frac{1}{2} \chi^2 \dot{\boldsymbol{\theta}}^2 - 2\Phi(\chi, \boldsymbol{\theta})$$

geometrical Shapiro

Geometrical optics

High frequency limit $\omega \rightarrow \infty$ (the classical limit $\hbar \rightarrow 0$)

$$F(\mathbf{r}, f) = \int D\theta \exp \left(i\omega \int_0^{\chi_s} L(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}}, \chi) d\chi \right)$$

Only paths satisfying $\frac{\delta t_d}{\delta \theta} = 0$ contribute. (**Fermat's principle**)

$$F(\omega) = \sqrt{\mu_1} + \sum_{j=2}^n \sqrt{\mu_j} \exp [i(\omega T_{1j} - \pi n_j)]$$

μ_1 : magnification of the first arrival image

$$T_{1j} = T_1 - T_j$$

Superluminal GWs?

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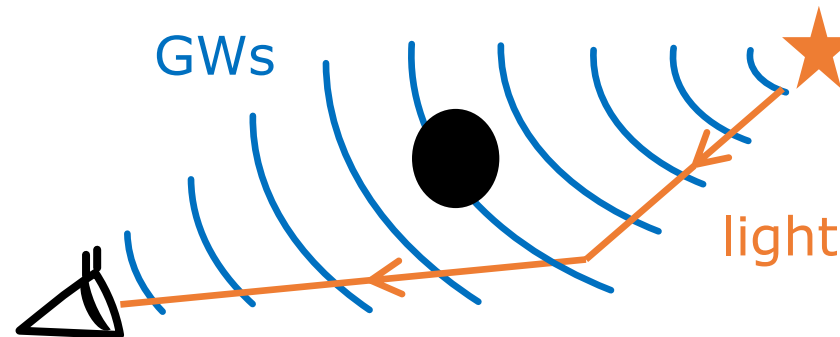
ARRIVAL TIME DIFFERENCES BETWEEN GRAVITATIONAL WAVES AND ELECTROMAGNETIC SIGNALS DUE TO GRAVITATIONAL LENSING

RYUICHI TAKAHASHI

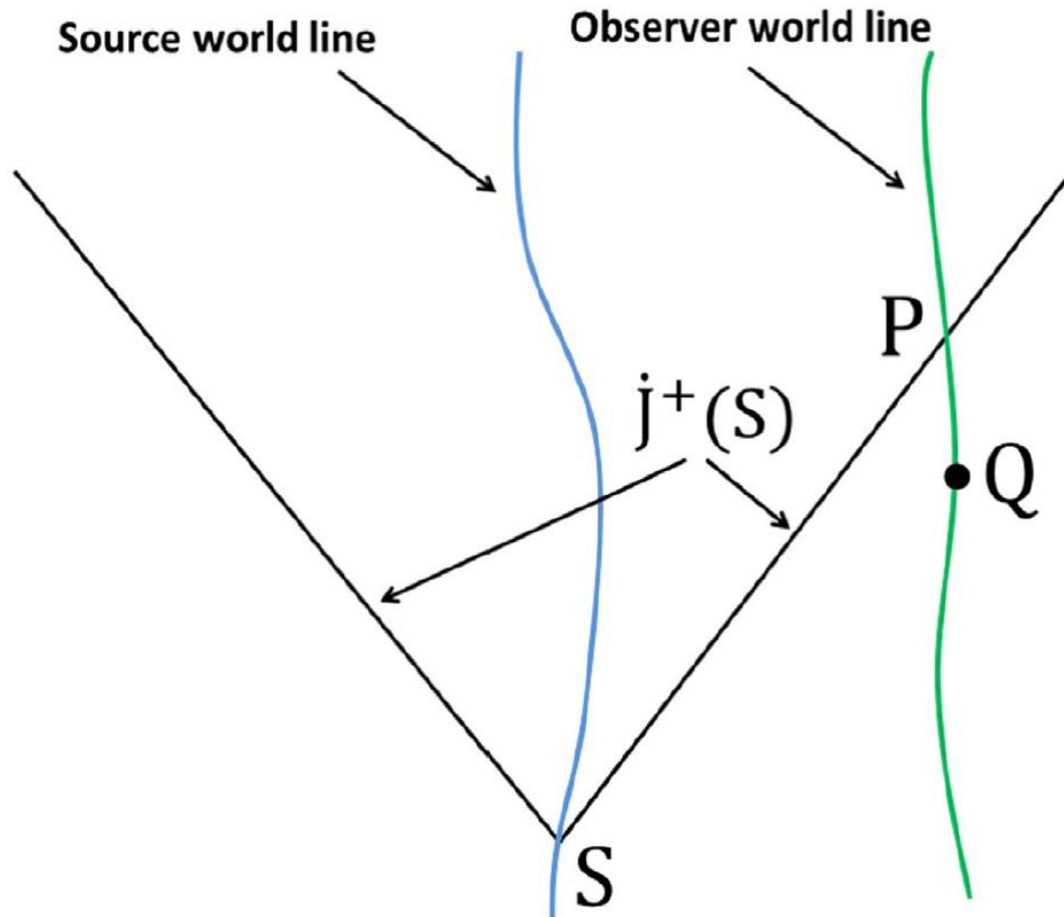
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ABSTRACT

In this study we demonstrate that general relativity predicts arrival time differences between gravitational wave (GW) and electromagnetic (EM) signals caused by the wave effects in gravitational lensing. The GW signals can arrive *earlier* than the EM signals in some cases if the GW/EM signals have passed through a lens, even if both signals were emitted simultaneously by a source. GW wavelengths are much larger than EM wavelengths; therefore, the propagation of the GWs does not follow the laws of geometrical optics, including the Shapiro time delay, if the lens mass is less than approximately $10^5 M_{\odot} (f/\text{Hz})^{-1}$, where f is the GW frequency. The arrival time difference can reach $\sim 0.1 \text{ s} (f/\text{Hz})^{-1}$ if the signals have passed by a lens of mass $\sim 8000 M_{\odot} (f/\text{Hz})^{-1}$ with the impact parameter smaller than the Einstein radius; therefore, it is more prominent for lower GW frequencies. For example, when a distant supermassive black hole binary (SMBHB) in a galactic center is lensed by an intervening



Phase velocity, group velocity are superluminal



Physical signals propagating from S to Q break causality



On Arrival Time Difference Between Lensed Gravitational Waves and Light

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Abstract

It is known that geometrical optics no longer applies to gravitational lensing if the wavelength of a propagating wave becomes comparable to or larger than the Schwarzschild radius of a lensing object. We investigate the propagation of gravitational waves in wave optics, particularly focusing on the difference between their arrival time and the arrival time of light. We argue that, contrary to the observation in the previous work, gravitational waves never arrive at an observer earlier than light when both gravitational waves and light are emitted from a same source simultaneously.

Unified Astronomy Thesaurus concepts: [Gravitational lensing \(670\)](#)

Lensed GWs never arrive earlier than light

Simple explanation

$$F(\omega) = \sum_{\text{path}:\boldsymbol{\theta}} e^{i\omega t_d[\boldsymbol{\theta}]}$$

$$h_L(t) = \int f(t - t') h_0(t') dt'$$

$$f(t) = \sum_{\boldsymbol{\theta}} \int \frac{d\omega}{2\pi} e^{i\omega t_d[\boldsymbol{\theta}] - i\omega t} = \sum_{\boldsymbol{\theta}} \delta(t - t_d[\boldsymbol{\theta}])$$

$$f(t) = 0 \quad \text{for } t < \min_{\boldsymbol{\theta}} t_d[\boldsymbol{\theta}] = T_{\min}$$

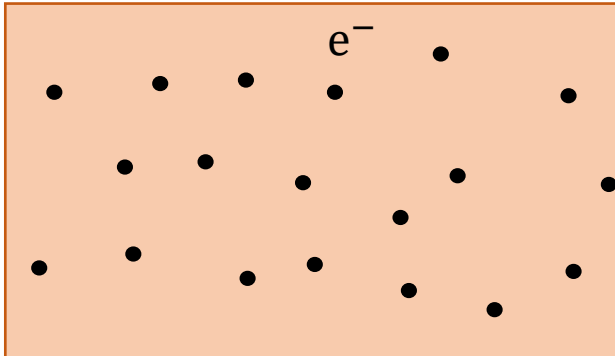
$T_{\min}$: travel time of geodesic (=light)

Lensed GWs never arrive earlier than light

Lensed signal comes only after its geometrical optics counterpart (remains inside the lightcone)

No violation of causality in GL

Kramers-Kronig (KK) relation

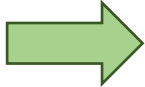


A diagram showing a collection of black dots representing electrons (e^-) within a rectangular region. To the right of the dots, the equation $D(\omega) = \epsilon(\omega)E(\omega)$ is written. Below this equation, two lines of text define the terms: $n (= \text{Re}\epsilon)$: refractive index and $\kappa (= \text{Im}\epsilon)$: extinction coefficient.

e^-

$D(\omega) = \epsilon(\omega)E(\omega)$

$n (= \text{Re}\epsilon)$: refractive index
 $\kappa (= \text{Im}\epsilon)$: extinction coefficient

Causality  $\epsilon(t) = 0$ for $t < 0$

KK relation

$$n(\omega) - 1 = \frac{1}{\pi} \oint d\omega' \frac{\kappa(\omega')}{\omega'(\omega' - \omega)}$$

principal value integral

KK relation is satisfied in any materials

K-K relation in GL

Causality  KK relation

$$\operatorname{Re} F(\omega) \equiv 1 + K(\omega),$$

$$\operatorname{Im} F(\omega) \equiv S(\omega).$$

GL version of KK relation

Tanaka and TS
PRD 108, 044015 (2023)

$$\frac{K(\omega)}{\omega} = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{d\omega'}{\omega' - \omega} \frac{S(\omega')}{\omega'},$$

$$\frac{S(\omega)}{\omega} = -\frac{1}{\pi} \int_{-\infty}^{\infty} \frac{d\omega'}{\omega' - \omega} \frac{K(\omega')}{\omega'}.$$

KK relation is satisfied in any lens

Derivation of the K-K relation

- Causality

$$\Longleftrightarrow F(\omega) = \int_{T_{min}}^{\infty} dt f(t) e^{i\omega t}$$

good convergence in I^+

$$\Longleftrightarrow \underbrace{F(\omega) e^{-i\omega T_{min}}}_{\substack{\text{redefine} \\ F(\omega)}} = \int_0^{\infty} dt f(t + T_{min}) e^{i\omega t}$$

F is **regular** in **upper half** of complex ω -plane (I^+)

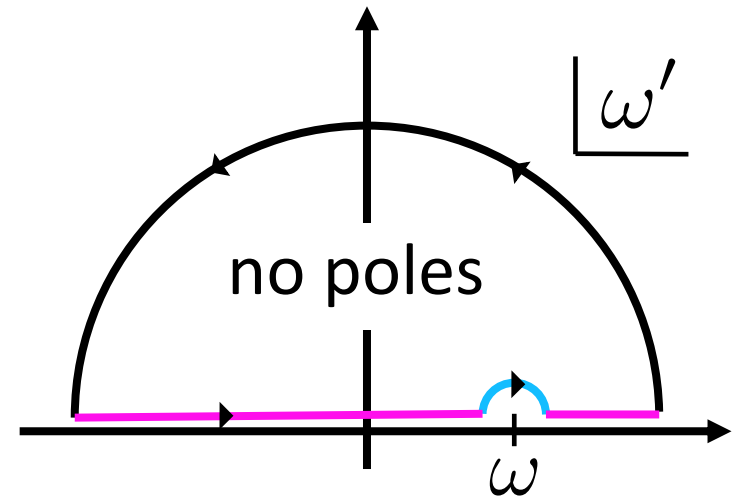
- $F(0) = 1, \quad \lim_{\omega \rightarrow \infty} |F(\omega)| < C$

$G(\omega) := \frac{F(\omega)-1}{\omega}$ has no poles on the real axis and in I^+ and $G \rightarrow 0$ ($\omega \rightarrow \infty$).

Derivation of the K-K relation

- Cauchy's integral theorem

$$\oint_C \frac{d\omega'}{\omega' - \omega} \frac{F(\omega') - 1}{\omega'} = 0$$



$$\Leftrightarrow \underbrace{\int_{-\infty}^{\infty} \frac{d\omega'}{\omega' - \omega} \frac{F(\omega') - 1}{\omega'}}_{\text{pink line}} - \underbrace{\pi i \frac{F(\omega) - 1}{\omega}}_{\text{blue line}} = 0$$

$$\Leftrightarrow \operatorname{Re} F(\omega) = 1 + \frac{\omega}{\pi} \int_{-\infty}^{\infty} \frac{d\omega'}{\omega' - \omega} \frac{\operatorname{Im} F(\omega')}{\omega'}$$

Sum rules (implications of the K-K relation)

Thomas-Reich-Kuhn (TRK) rule (Condensed matter)

$$\int_0^\infty d\omega \kappa(\omega) = \frac{\pi}{2} \omega_p^2 \quad \omega_p: \text{plasma mass}$$

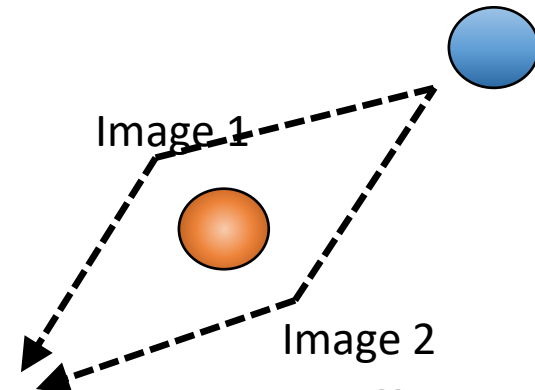
GL of GWs

$$\frac{2}{\pi} \int_0^\infty d\omega \frac{S(\omega)}{\omega} = 1 - \sqrt{\mu_1}$$

Geometrical optics

$$F(\omega) = \sqrt{\mu_1} + \sum_{j=2}^n \sqrt{\mu_j} \exp[i(\omega T_{1j} - \pi n_j)]$$

μ_1 : magnification of the first arrival image



$$F(\omega) = |F(\omega)|e^{i\theta(\omega)}$$

The magnitude and the phase have clear physical meaning

Does KK relation for $|F|$ and θ exist?

※ **YES** for condensed matter

e.g. Stern 1963

$$\ln F(\omega) = \ln |F(\omega)| + i\theta(\omega)$$

Physics experiment class (undergraduate, Science Tokyo)

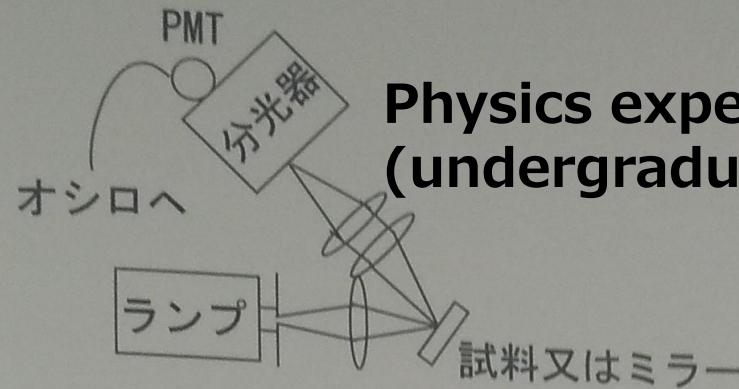


図 18: 反射測定 of 配置図

多くの物質は吸収スペクトルが測定できないため、一般には垂直反射の反射スペクトル $R(\omega) (= |r(\omega)|^2)$ の測定から誘電率を求めることが多い。図のような測定系を組み立て、試料からの反射強度 $I_R(\omega)$ とアルミミラーからの反射強度 $I_{R0}(\omega)$ を測定する。出来るだけ垂直反射の条件に近づけるため、入射角は小さいほうが良い。反射率 $R(\omega)$ は、 $R(\omega) = I_R(\omega) / I_{R0}(\omega)$ から求める。ここではアルミミラーの反射率を 1 であると近似する。垂直反射の場合の振幅反射率 $r(\omega)$ はフレネルの反射公式 (物理学実験第一”光の波動的性質”参照) より、

$$r(\omega) = \frac{n + i\kappa - 1}{n + i\kappa + 1} = \sqrt{R(\omega)} \exp[i\theta(\omega)] \quad (91)$$

または、

$$\ln r(\omega) = \ln \sqrt{R(\omega)} + i\theta(\omega). \quad (92)$$

ここで、 θ は反射の際の光の位相変化を示す。クラマース・クローニツヒの関係¹⁴ より、

$$\theta(\omega) = \frac{2\omega}{\pi} P \int_0^\infty \frac{\ln \sqrt{R(\nu)}}{\nu^2 - \omega^2} d\nu. \quad (96)$$

したがって、反射スペクトルが十分広い光エネルギー領域で測定できれば、(96) 式より $\theta(\omega)$ を求めることができる。(91) 式より、屈折率の実部と虚部は、

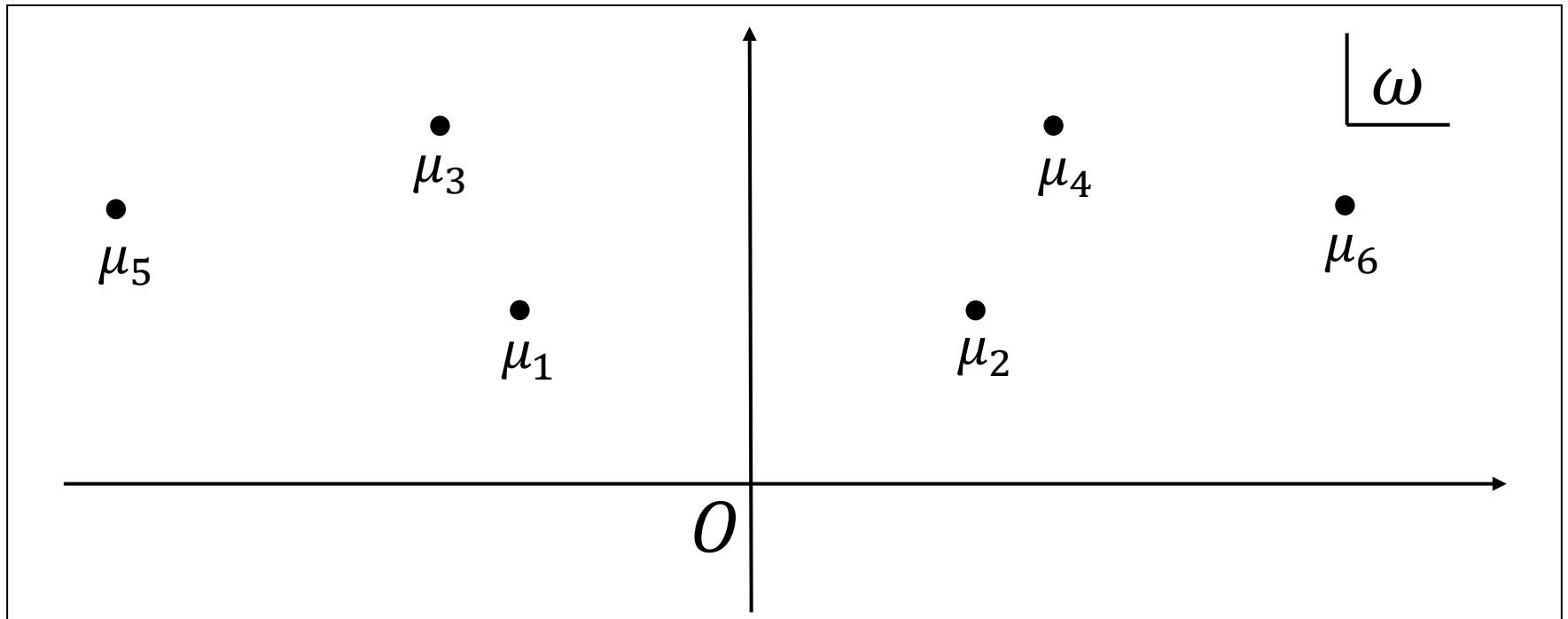
General theory

$A(\omega)$: analytic in upper-half plane

$A_{new}(\omega) = B(\omega)A(\omega)$ is also analytic

$$B(\omega) = \prod_n \left(\frac{\omega - \mu_n}{\mu_n^* - \omega} \right)$$

Blaschke product



$$|B(\omega)| = 1 \text{ for } \omega \in \mathbb{R}$$

$$|A_{new}(\omega)| = |A(\omega)| \text{ for } \omega \in \mathbb{R}$$

$|A(\omega)|$ does not uniquely determine the phase of $A(\omega)$.

Condensed matter physics

Stern 1963

$$\theta(\omega) = (2\omega/\pi) P \int_0^\infty \ln [r(\omega')] (\omega^2 - \omega'^2)^{-1} d\omega' + \xi(\omega), \quad (20.5)$$

$$\theta(\omega) \rightarrow \pi \text{ for } \omega \rightarrow \infty$$

$$r(\omega) \rightarrow \frac{\omega_P^2}{2\omega^2} \text{ for } \omega \rightarrow \infty$$



$$\xi(\omega) = 0 \quad (\text{Blaschke product is 1})$$

Possible existence of the B-product in GL

$$\theta(\omega) = -\frac{2\omega}{\pi} \int_0^\infty \frac{\ln |F(\omega')|}{\omega'^2 - \omega^2} d\omega' - i \sum_n \ln \left(\frac{\omega - \mu_n}{\mu_n^* - \omega} \right)$$



$$\omega \rightarrow \infty$$

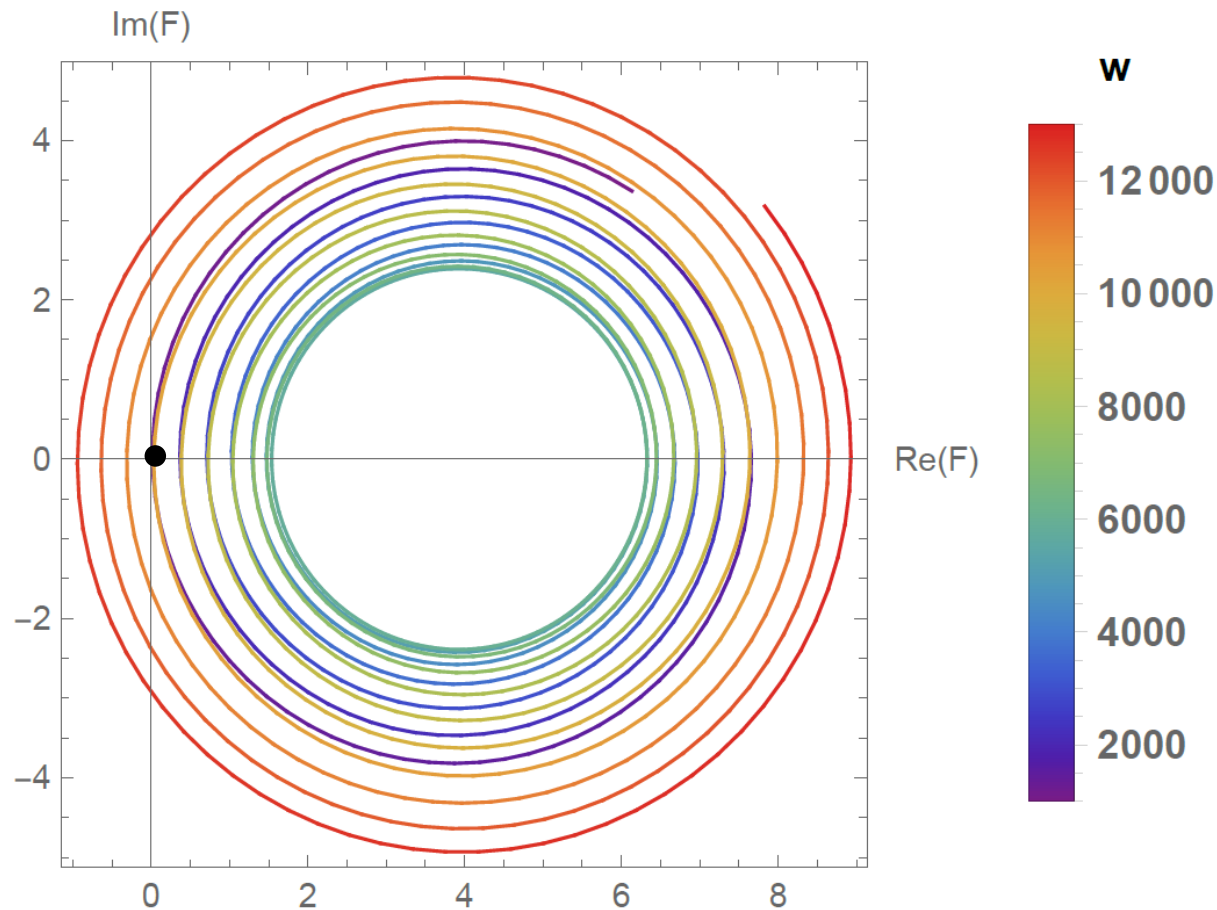
(geometrical optics regime)

$$\begin{aligned} & \arg(\sqrt{\mu_1} + \sum_{j=2}^N \sqrt{\mu_j} e^{i\omega \Delta t_j - \pi i n_j \operatorname{sgn}(\omega)}) \\ &= -\frac{\omega}{\pi} \int_0^\infty \ln \left(\sum_{j=1}^N \mu_j + 2 \sum_{j>k}^N \sqrt{\mu_j \mu_k} \cos(\omega' \Delta t_{jk} - \pi n_{jk}) \right) \frac{d\omega'}{\omega'^2 - \omega^2} - i \ln B(\omega) \end{aligned}$$

Bounded for $\omega \rightarrow \infty$

Possible existence of the B-product in GL

Lens = NFW profile $(\mu_1, \mu_2, \mu_3) = (15.40, 17.02, 3.02)$



The Blaschke product exists in this case

KK Relations for the magnitude and the phase

$$\theta(\omega) = -\frac{2\omega}{\pi} \int_0^\infty \frac{\ln |F(\omega')|}{\omega'^2 - \omega^2} d\omega' - i \sum_n \ln \left(\frac{\omega - \mu_n}{\mu_n^* - \omega} \right)$$

$$\ln |F(\omega)| = \frac{2}{\pi} \int_0^\infty \left(\frac{\omega'}{\omega'^2 - \omega^2} - \frac{1}{\omega'} \right) \theta(\omega') d\omega'$$

The phase is not uniquely determined from the magnitude.

The magnitude is uniquely determined from the phase.

Low-frequency behavior

Choi et al. 2021, Tambalo et al. 2023

$$F(\omega) = \begin{cases} 1 + 2^{-\frac{k}{2}} e^{-i\frac{k\pi}{4}} \Gamma\left(1 - \frac{k}{2}\right) \omega^{\frac{k}{2}} + \dots & \text{(Generalized SIS)} \\ 1 + \frac{w}{2} (\pi + 2i \ln w) + \dots & \text{(Point - mass lens)} \end{cases} \quad \rho(r) \propto \frac{1}{r^{k+1}}$$

Now assume $|F(\omega)| = 1 + A\omega^\alpha + \dots$

Point mass ($\alpha = 1$), SIS ($\alpha = \frac{1}{2}$), gSIS ($\alpha = \frac{k}{2}$)

KK relation


$$\theta(\omega) = \begin{cases} -\tan\left(\frac{\pi\alpha}{2}\right) \ln |F(\omega)| + \dots & 0 \leq \alpha < 1 \\ \ln |F(\omega)| \ln \omega + \dots & \alpha = 1 \end{cases}$$

Causality strongly restricts the low-frequency behavior of the lens signal.

Application to lensing measurement

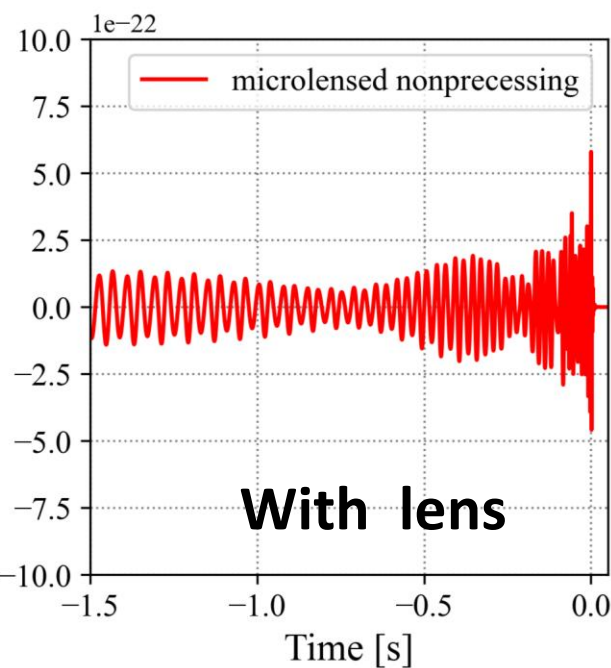
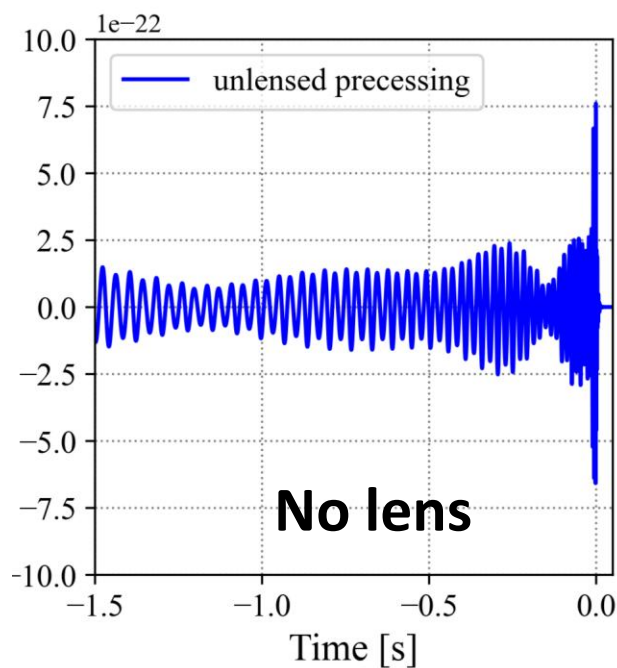
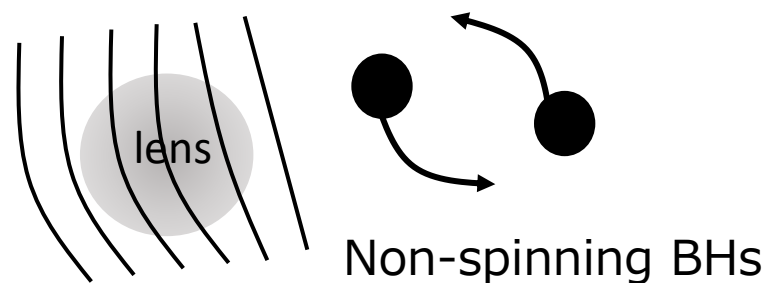
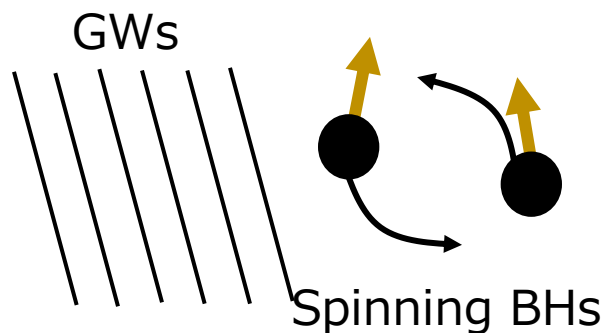
Tanaka et al. 2025

$$\mathbf{h}_L(\mathbf{f}) = F(\mathbf{f})h_0(\mathbf{f})$$

How can we know it is lensed or unlensed?

(This question persists even if the GW data is free from errors)

Lensing of GWs



Kim&Liu 2023

Some GW sources mimic lens signal

Application to lensing measurement

Tanaka, Prabhu, Kapadia, TS (2025)

Lensing hypothesis

$h_{obs}(f)$ is a waveform lensed from template $h_0(f; \theta_s)$

$$F_{obs}(f) := \frac{h_{obs}(f)}{h_0(f; \theta_s)}$$

Any $F_{obs}(f)$ which does not satisfy the KK relation is ruled out as **false** GL signal.

***This is independent of the lensing model.**

Can the KK relation be used to reject the hypothesis even when the frequency band is finite?

Application to lensing measurement

Quantification of the violation of the KK

$$\Delta := (LHS) - (RHS)$$

Assumptions on measurements:

1. $\omega_{min} \ll \omega \ll \omega_{max}$, 2. wave optics regime at ω_{min} , 3. no noise

If $F(f)$ is a correct lens signal,

$$\text{Re}\Delta = O(1) \times K(\omega_{min})$$

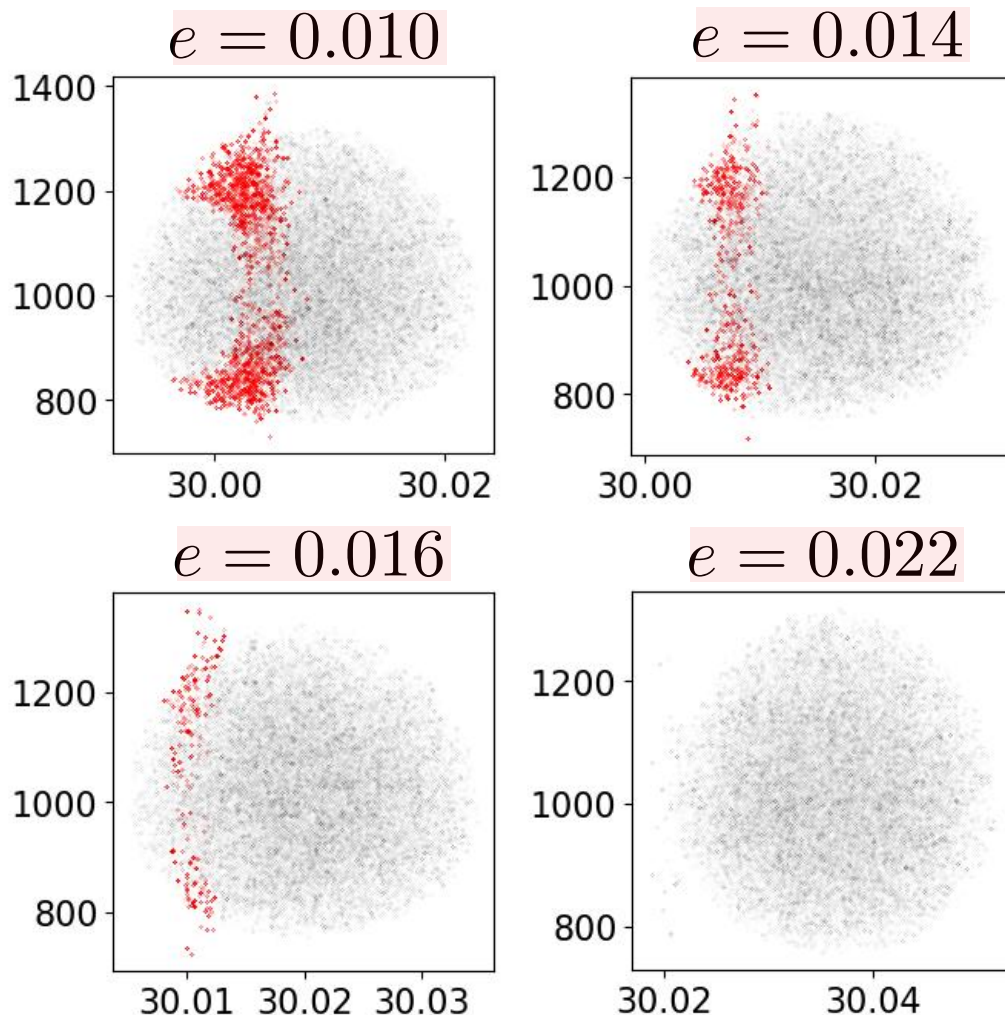
We reject unlensed template if $\text{Re}\Delta > r_{th} K(\omega_{min})$

Proof of concept

Tanaka, Prabhu, Kapadia, TS (2025)

GWs from an eccentric orbit

■ $\phi_{obs} = \phi_{ecc}$ $m - d$ plot



ϕ_{ecc} 's parameters

$$m = 30M_{\odot}$$

$$d = 1000 \text{ Mpc}$$

$$t_{coal} = 0$$

$$\varphi_{coal} = 0$$

● ruled out

● not ruled out

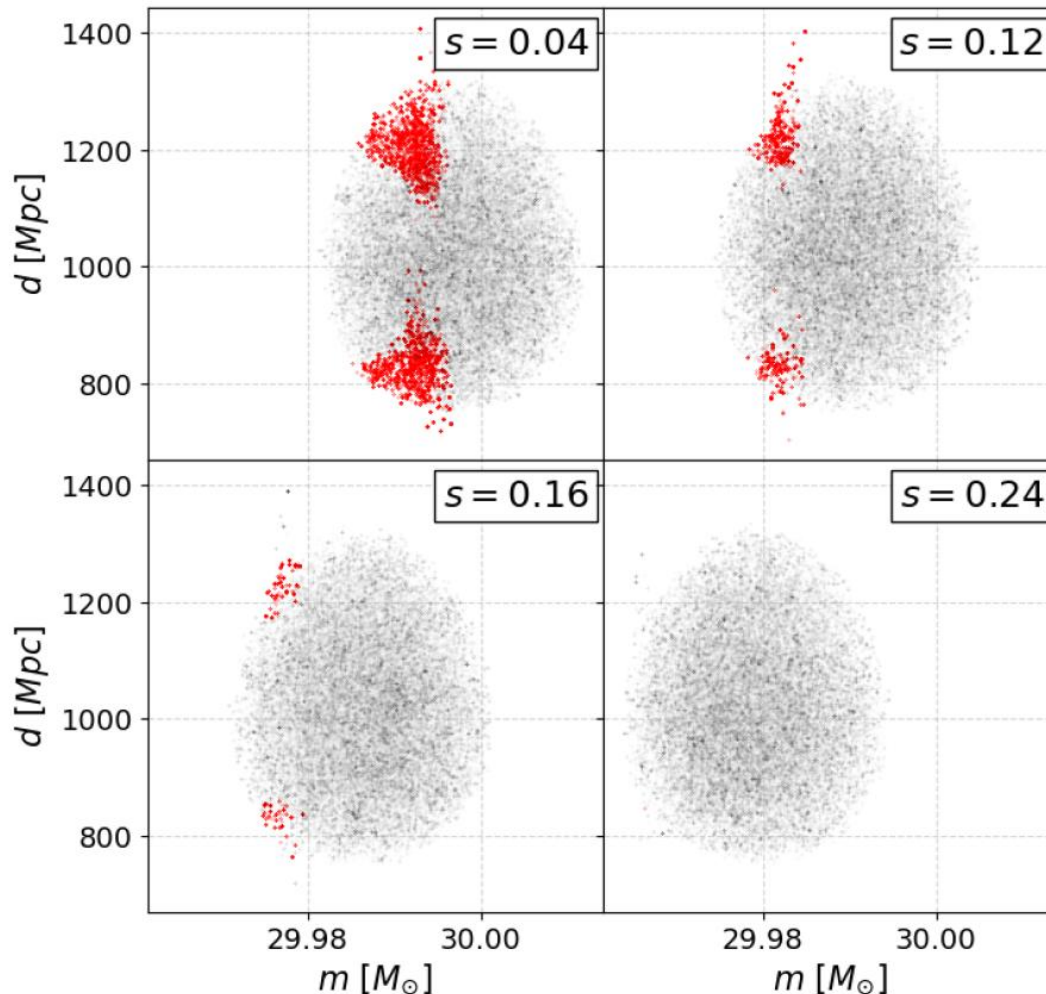
Large eccentric orbit can be distinguished from lensing

Proof of concept

Tanaka, Prabhu, Kapadia, TS (2025)

GWs from a spinning orbit

■ $\phi_{obs} = \phi_{spin}$ $m - d$ plot



parameters

$$m = 30M_\odot$$

$$d = 1000 \text{ Mpc}$$

$$t_{coal} = 0$$

$$\varphi_{coal} = 0$$

● ruled out

● not ruled out

Orbit with large spin can be distinguished from lensing

Summary

Gravitational lensing is indispensable tool in cosmology.

GL signal obeys the Kramers-Kronig relation.

KK relations provide a model-independent diagnostic for gravitational-wave lensing